

Conclusive Proposition — From Cloud to Clock: A Geometric Law Governs the Positions of n , p , and q

1. Origin of the Approach: Dyadic Concentric Circles

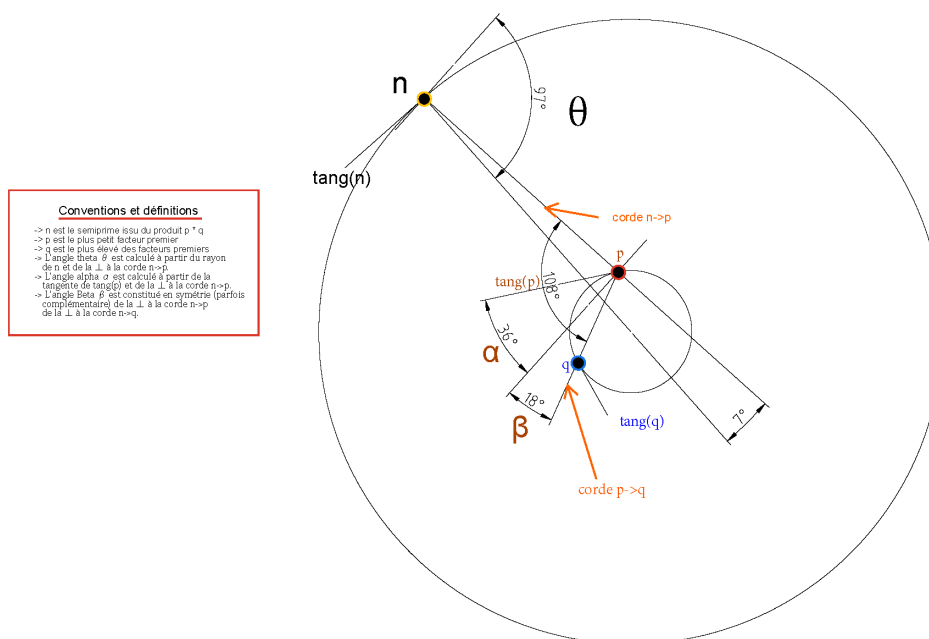


Figure 1 Preliminary intuition guiding this research.

Initially, our experimental setup resembled a clock without a dial: a hand (the semiprime n) rotated on its dyadic level, but nothing indicated where the "hours" were or how to read the position of p (and then q) from this single position. We had visible kinematics—clouds, correlations, shapes—but no **graduation** capable of converting this geometry into usable arithmetic information.

The guiding idea, consistent with the dyadic perspective we established in the December 26 note (stabilization of behaviors at large scales, palier invariance, "structural" rather than purely probabilistic reading), was that dyadic levels constitute a natural coordinate space. It took me a few weeks to shift from a horizontal representation to the idea of **concentric circles** sharing a common center at 0, where each level K serves as a stable "dial" beyond a certain threshold (empirically, $K \geq 20$).

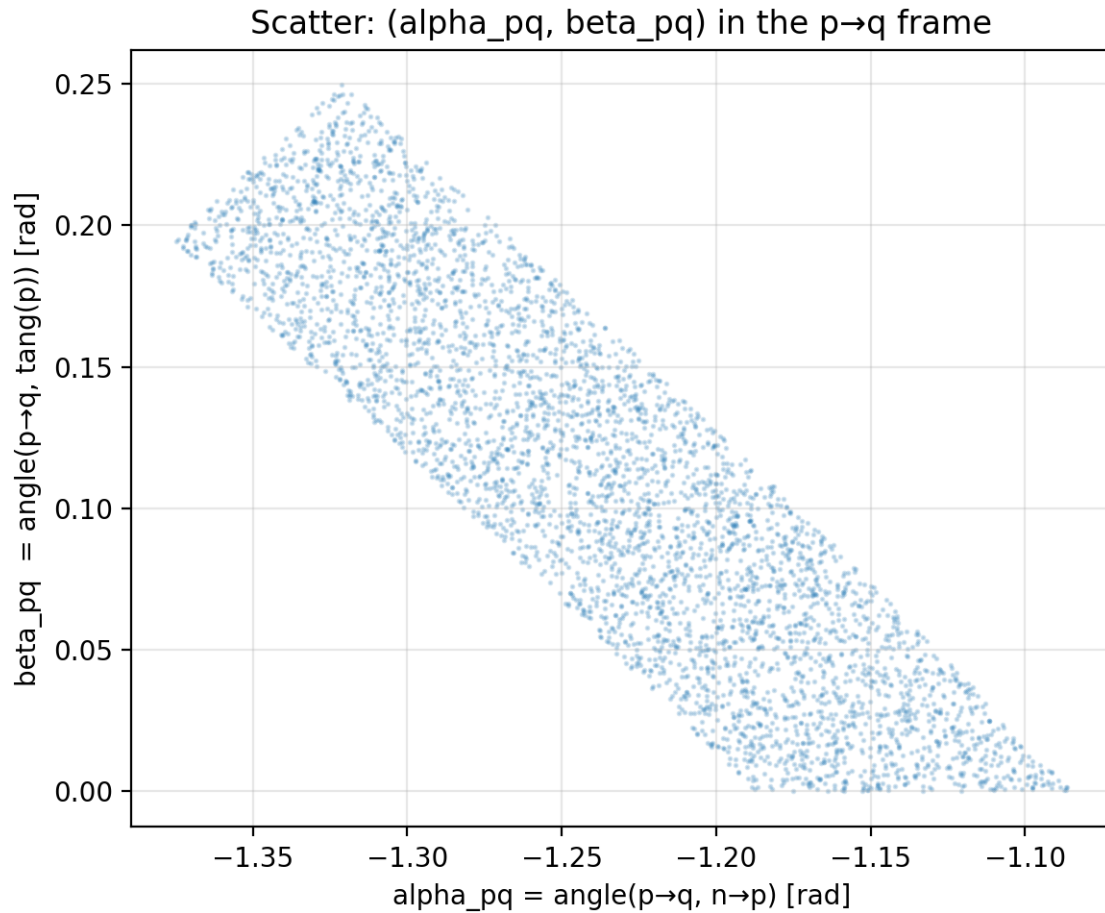
Ce travail préliminaire a fait l'objet du dépôt github initial accessible via ce lien:

[DYADIC STRUCTURAL PERSPECTIVE](#)

The successive developments and explorations leading to the present announcement are grouped in a later repository:

[DYADIC PHASE TRANSPORT IN SEMIPRIMES INTEGERS](#), based on the idea of geometric conservation in nof its relationship with p and q , hence the pendular relation approach.

2. The Initial Observation: An Admissible but Unexploitable Cloud



In the initial cloud (e.g., θ_p as a function of θ_n , or β as a function of α), we already observed a clear slope: a sign that a mechanism exists, but **mixed** with multiple regimes (carry, geometric admissibility, permutations, etc.). In other words: a clock, yes—but with all the hands stacked.

The first "dial" breakthrough consisted of **discretizing** the angular space of θ_n (bins) and measuring, bin by bin, the q_{95} width of θ_p . A decisive fact emerged: *certain sectors of θ_n concentrate θ_p in a much narrower corridor than the global distribution (high gains)*. We still couldn't read the time, but we discovered the first relevant graduations.

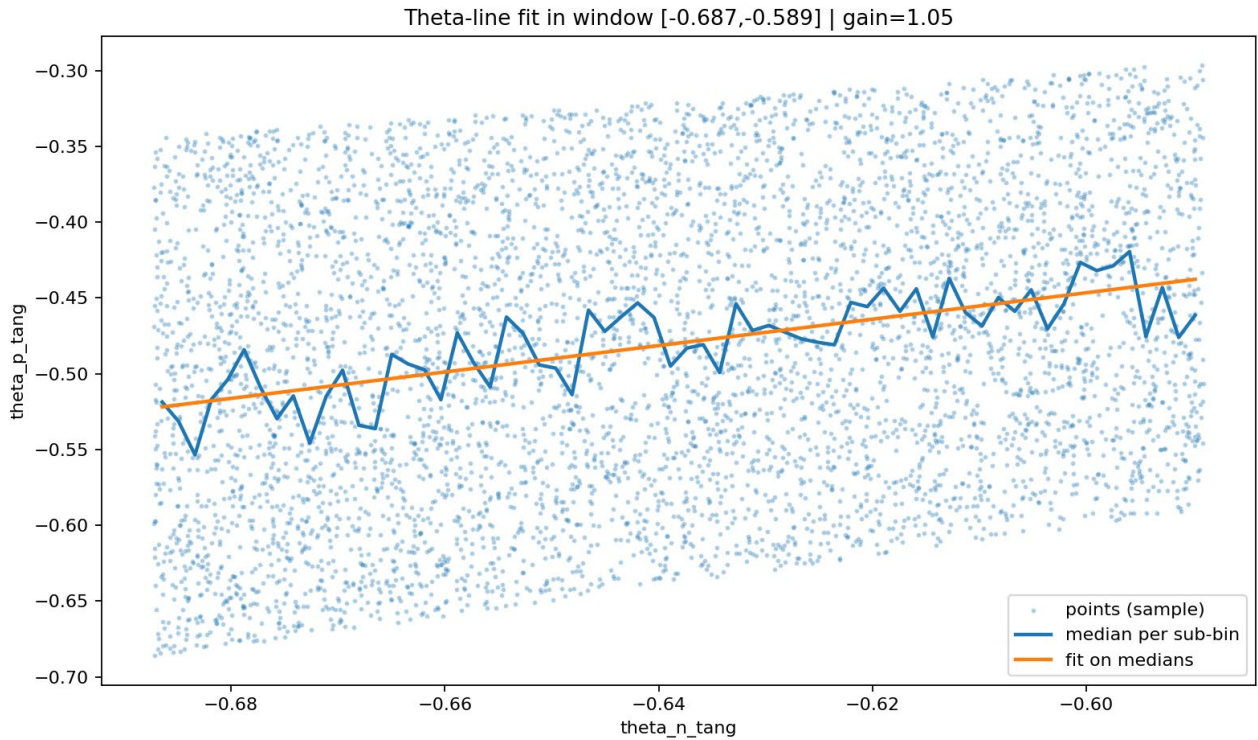


Figure 2 In the initial cloud (e.g., θ_p as a function of θ_n , or β as a function of α), we already observed a clear slope: a sign that a mechanism exists, but **mixed** with multiple regimes (carry, geometric admissibility, permutations, etc.). In other words: a clock, yes—but with all the hands stacked. The first "dial" breakthrough consisted of **discretizing** the angular space of θ_n (bins) and measuring, bin by bin, the q_{95} width of θ_p . A decisive fact emerged: certain sectors of θ_n concentrate θ_p in a much narrower corridor than the global distribution (high gains). We still couldn't read the time, but we discovered the first relevant graduations.

At this stage, the clock remained incomplete: we had a hand (θ_n) and favorable sectors, but we lacked a **fine mechanism** to transform the residual dispersion into a quasi-deterministic trajectory.

This is where $\Delta_{pq} = \text{wrap}(\theta_p - \theta_q)$ naturally took on the role I wanted to define as analogous to that of the second hand of a watch, providing the linking mechanism with an additional precision spring that encodes a relative phase between the factors, thus an **internal degree of freedom** for the pair (p, q) .

Empirically, by binning Δ_{pq} , we obtained a clear narrowing of the θ_p distribution (gains > 3 in the first slices), confirming that Δ_{pq} carries usable directional information, though inoperable at high dyadic levels.

We therefore sought to move from the raw cloud to an instrumental reading by using all the geometric links underlying what we felt proceeded from the intimate cohesion of the triplet $\{n, p, q\}$ during its evolution in their respective circumferences.

3. Kinematic Refinement: From Parallelepiped to Line

Progressive refinement (conditioning by dyadic level, carry, angular subdivisions) reveals a decisive fact:

- The initial cloud is not homogeneous;
- It is the superposition of **local kinematic regimes**.

When these regimes are isolated, the parallelepiped **tightens** and reveals:

- An **affine structure**,
- Materialized by a median line,
- Around which dispersion becomes secondary.

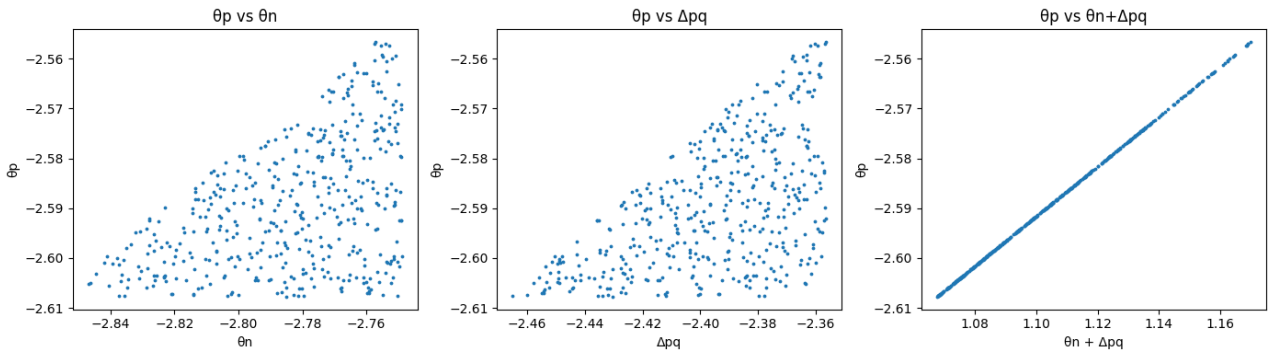
In other words: what appeared as a volumetric admissible zone turns out to be, in reality, a **quasi-linear transport**.

The intuition we pursued proved correct:

As shown in the final figure (opening Section 4 of this document), produced from 500,000 triplets, we moved from a parallelepiped to an affine line, and this line encodes the kinematics of the transfer from n to p .

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4. The Structural Revelation: The Law of Angular Transport



The joint analysis of θ_n and Δ_{pq} leads to the robust empirical relation:

$$\theta_p \approx \text{wrap}(\theta_n + \Delta_{pq}) + \varepsilon$$

In this framework:

- θ_n sets the **global position on the dyadic dial**,
- Δ_{pq} acts as an **internal spring**, a *second hand* providing fine resolution,
- θ_p becomes a **determined reading**, locatable with unexpected precision.

What the figure shows unambiguously is that while θ_p vs. θ_n and θ_p vs. Δ_{pq} remain cloudy, θ_p vs. $(a \cdot \theta_n + b \cdot \Delta_{pq})$ aligns **almost perfectly on a line**.

This is no longer a correlation: It is an **internal kinematic law** of the semiprime.

From n alone:

1. Calculate θ_n ,
2. Sweep Δ_{pq} in **very coarse windows**,
3. Each Δ_{pq} gives:

$$\theta_p \approx \theta_n + \Delta_{pq}$$

4. Hence, **an ultra-narrow interval for p** ,
5. Arithmetic return: test candidate p values.

The gain is no longer $\times 5$ or $\times 10$.

Here, we see **$\times 40$ – $\times 50$ locally**, and this is not an isolated case.

5. Scope of the Result: Unexpected Precision at 1024 Bits

The essential—and unprecedented—point is as follows:

This law appears **at 1024 bits**, without direct numerical approximation or heuristics from classical factorization.

The observed uncertainty reduction (high gain factors on the admissible angular width of θ_p) shows that:

- The search space for p is not isotropic,
- It is **guided by an internal geometry**,
- Exploitable independently of GNFS methods.

This is not yet a factorization algorithm, but a **structured reduction of the space**, based on a real geometric law.

The next (logical, not speculative) step is to:

A) Formalize the Law

For each regime:

$$\theta_p = a(\text{bin}) \cdot (\theta_n + \Delta_{pq}) + b(\text{bin})$$

with $a \approx 1$

B) Map the Regimes

- Indexed by $(\text{bin } \theta_n, \text{bin } \Delta_{pq}, \text{carry})$,
- This is the heatmap $\rightarrow$ **transport table**.

C) Implement Factorization

A hybrid algorithm:

- Geometry $\rightarrow$ reduction,

- Arithmetic → validation.

6. Open Consequences

This **spatial revelation** suggests several major consequences that may extend beyond the specific field of my focus:

Conceptual Semiprimes possess an **internal deterministic kinematics**, absent from purely arithmetic models.

Methodological The dyadic concentric representation constitutes a **tool for structural exploration**, not just visualization.

Cryptographic Any security method based on complete ignorance of the position of p must be reevaluated if such angular localization is exploitable.

Theoretical Factorization can be reformulated as a **geometric reading problem**, not exclusively a computational problem.

7. Conclusion

This work shows that by changing our perspective—which I have continually done throughout this study by placing integers on circles, accepting their phase, and listening to their kinematics—what was once a cloud becomes a clock.

And once the clock is revealed, the question is no longer *whether* it tells the time, but **what we will now do with it**. Even as a mere layperson, it seems to me that this approach opens real perspectives beyond what it says about my own insignificant self.